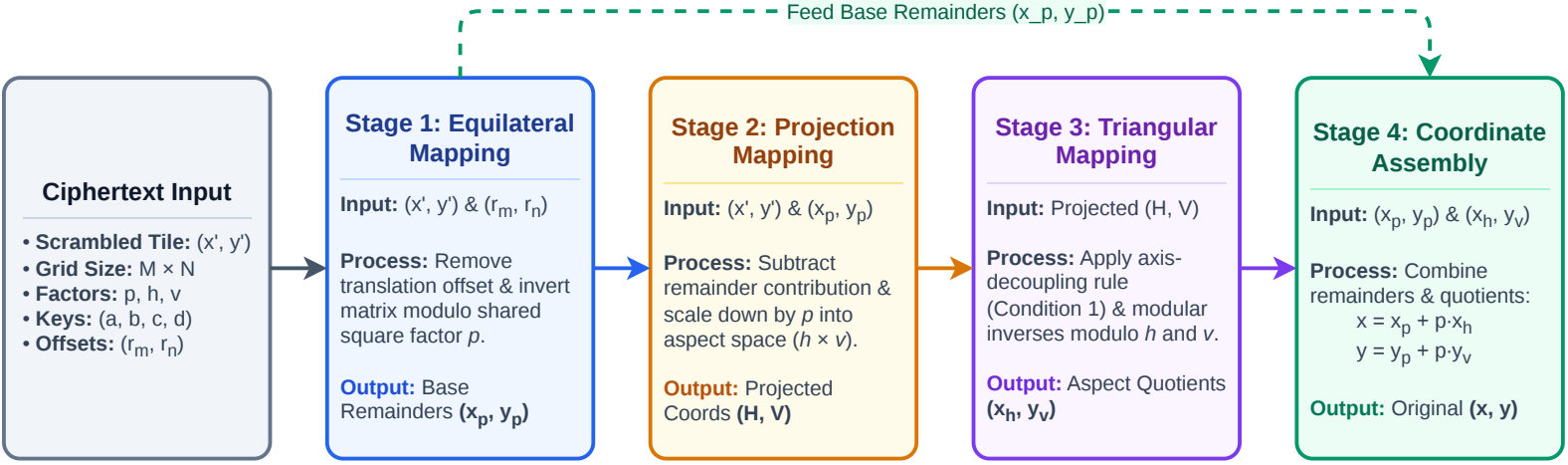

Single-Pass O(1) Coordinate Recovery per Tile — Precomputed into 1D Lookup Tables (*igrt_client.js*) for **0.32–0.51 ms** Canvas rendering.